Code-Layout, Readability and Reusability

Date	29 June 2026
Team ID	
Project Name	Rising Waters-ML FLood Prediction Model
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/ Partial)	Remarks
1	Consistent Indentation	Proper indentation maintained throughout Python, HTML, CSS and JavaScript files.	Yes	Improves readability and maintenance.
2	Proper File Structure	Project organized into Templates, Static, Dataset, Notebook, Models, Reports and Screenshots folders.	Yes	Well-structured directory layout following Flask architecture.
3	Meaningful Variable Names	Variables use descriptive names related to rainfall, prediction and weather parameters.	Yes	Makes the code easier to understand.
4	Function / Method Names	Functions are named according to their purpose such as prediction, preprocessing and routing.	Yes	Improves code clarity and maintainability.
5	Code Comments	Important sections contain comments explaining preprocessing, model loading and prediction logic.	Partial	Additional comments can be added for better documentation.
6	Modular Design	Frontend, backend, model files and datasets are separated into different modules.	Yes	Promotes maintainability and scalability.
7	No Redundant Code	CSS and JavaScript are separated from HTML to reduce duplication.	Yes	Code reuse is achieved across pages.
8	Error Handling	Basic validation is implemented for prediction requests and model loading.	Partial	More comprehensive exception handling can be added.

Reusable Components / Modules

S.No	Component/Module Name	Language / Technology	Reusability Level
1	Flask Backend (app.py)	Python, Flask	High
2	Machine Learning Model (floods.save)	XGBoost, Joblib	High
3	Data Transformer (transform.save)	Scikit-learn	High
4	HTML Templates	HTML5, Jinja2	High
5	CSS Stylesheets	CSS3	High
6	JavaScript Files	JavaScript	High
7	Dataset	Excel (.xlsx)	Medium
8	Jupyter Notebook	Python, Jupyter	Medium
9	Prediction Pages (chance.html, no_chance.html)	HTML, CSS, JavaScript	High
10	Images and Icons	PNG	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Project Structure	5	Organized according to Flask project standards with separate folders for frontend, backend and models.
Readability	5	Meaningful naming conventions and proper formatting make the code easy to understand.
Maintainability	5	Modular architecture allows individual components to be updated independently.
ability	5	CSS, JavaScript, HTML templates and machine learning model can be reused in future projects.
Documentation	4	Basic comments are included; additional inline documentation can further improve clarity.
Error Handling	4	Basic validation exists, but more detailed exception handling can be implemented.
Overall Code Quality	5	The project follows good coding practices with a clean structure, reusable modules and maintainable design.